

10. Recidivism Recidivism refers to the return to prison of a prisoner who has been released or paroled. The data that follow reports the group median age at which a prisoner was released from a federal prison and the percentage of those arrested for another crime.⁷ Use the *MS Excel* printout to answer the questions that follow.

Group Median Age (x)	22	27	32	37	42	47	52
% Arrested (y)	64.7	59.3	52.9	48.6	44.5	37.7	23.5

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.9779
R Square	0.9564
Adjusted R Square	0.9477
Standard Error	3.1622
Observations	7.0000

ANOVA

	df	SS	MS	Significance F
Regression	1	1096.251	1096.251	109.631
Residual	5	49.997	9.999	0.000
Total	6	1146.249		

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	93.617	4.581	20.436	0.000	81.842	105.393
x	-1.251	0.120	-10.471	0.000	-1.559	-

- Find the least-squares line relating the percentage arrested to the group median age.
- Do the data provide sufficient evidence to indicate that x and y are linearly related? Test using the t statistic at the 5% level of significance.
- Construct a 95% confidence interval for the slope of the line.
- Find the coefficient of determination and interpret its significance.

$$d. R^2 = 0.9564$$

⇒ model 可解釋 95.64% 的資料變異比例

⇒ model effective

$$a. \hat{y} = 93.617 - 1.251x$$

$$b. H_0: \beta = 0, H_a: \beta \neq 0$$

$$\alpha = 0.05$$

$$t_{STAT} = \frac{b}{\sqrt{\frac{MSE}{S_{xx}}}} \sim t_{\alpha/2, n-2}$$

$$t^* = -10.471$$

$$RR = \{t \mid |t| > t_{0.025, 5}\}$$

$$= \{t \mid |t| > 2.571\}$$

$$|-10.471| > 2.571 \Rightarrow \text{Reject } H_0$$

⇒ Data provide sufficient evidence to show that x and y are linearly related

$$c. 95\% \text{ CI}$$

$$b \pm t_{0.025, n-2} \sqrt{\frac{MSE}{S_{xx}}}$$

$$= -1.251 \pm 2.571 \times 0.12$$

$$= (-1.559, -0.992)$$